

The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL 1 MODULE, AUTUMN SEMESTER 2018-2019

**SYSTEMS AND ARCHITECTURE
(COMP 1030)**

Time allowed ONE Hour

*Candidates must **NOT** start writing their answers until told to do so*

Answer TWO questions out of THREE

Only silent, self-contained calculators with a single line display are permitted in this examination.

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Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

1. Networks (25 Marks)

- (a) Differentiate between the following terms:
- (i) Thicknet and thinnet
 - (ii) Simplex and full duplex communications
 - (iii) Broadband and baseband coaxial cables
 - (iv) Bit rate and baud rate [8 Marks]
- (b) With the help of diagrams, briefly explain the operations of a packet switching network using a datagram approach. List and discuss any one advantage and one disadvantage. [6 Marks]
- (c) Distinguish between adaptive and non-adaptive routing algorithms. Why is adaptive routing superior to non-adaptive routing? [5 Marks]
- (d) Define the term "flooding". [2 Mark]
- (e) Briefly explain the operations of the two (2) major types of bridges. [4 Marks]

2. Binary arithmetic, Processor Registers and Addressing modes (25 Marks)

- (a) An 8-bit register contains the value 01111011. Write the value of the register after completing the following shift operations in sequence.

- (i) SHR
- (ii) CIR
- (iii) A SHR
- (iv) SHR

[4 Marks]

- (b) Represent the number 12.125_{10} by using the given single and double precision formats given below.

Single precision format

1	8 bits	23 bits
Sign bit	Exponent in excess 127	Mantissa fraction

Double precision format

1	11 bits	52 bits
Sign bit	Exponent in excess 1023	Mantissa fraction

[8

Marks]

- (c) Explain the function of the following registers in a CPU.

- (i) PC
- (ii) IR
- (iii) MAR
- (iv) MBR
- (v) ACC

[10 Marks]

- (d) What is the difference between the direct and indirect addressing technique? How many references to memory are needed for each form of address mode to bring an operand into a processor register?

[3 Marks]

3. Interrupts, I/O and Instruction Pipelining (25 Marks)

- (a) Briefly describe the term "Instruction pipelining". [2 Marks]
- (b) What do you understand by the term "data dependency"? [2 Marks]
- (c) With the help of examples, briefly describe the three classes of data dependencies that could occur during program execution. [6 Marks]
- (d) When a multiple device interrupts have occurred, how does the processor decide which one to process? [8 Marks]
- (e) When DMA module takes control of a bus, and while it remains control of the bus, what does the processor do? [3 Marks]
- (f) List any four characteristics of "RISC" architectures. [4 Marks]